Town Planning

- The **growth of civilization has also resulted in more and more laws, regulations, and restrictions** and of course their violations as well.
- There are laws and laws, all around such as **constitutional, civic laws, traffic laws, financial laws, personal laws etc.**
- while **some of the laws are essential for the humanity, there are other which held ion the proper and planned growth of civilization in general**, in the olden days human beings had been wandering from place to place in search of livelihood but the modernization have put an end to all such adventures.

Town Planning





Town Planning

- Knowledge, have developed lot of conveniences for their proper living.
- **Human beings no longer have to roam around for proper shelter but live in better planned and construction of such houses**, same basic rules and regulations which may be termed as **„building Bye-laws are absolutely necessary.**

Town Planning

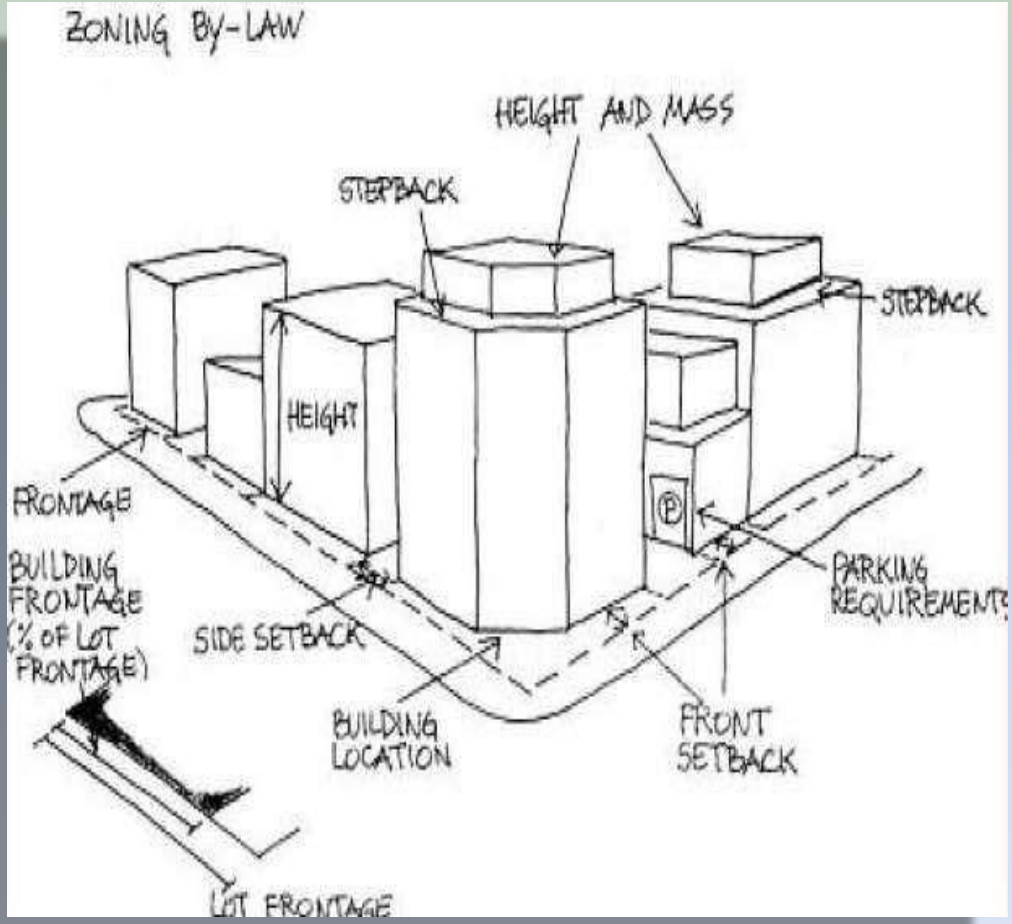




Town Planning

- **Building Bye-Laws are tools used to regulate coverage, height, building bulk, and architectural design and construction aspects of buildings so as to achieve orderly development of an area.** They are mandatory in nature and serve to protect buildings against fire, earthquake, noise, structural failures and other hazards.
- **In India, there are still many small and medium sized towns which do not have building bye-laws and in the absence of any regulatory mechanism**, such towns are confronted with excessive coverage, encroachment and haphazard development resulting in chaotic conditions, inconvenience for the users, and disregard for building aesthetics, etc

Building Bye -Laws





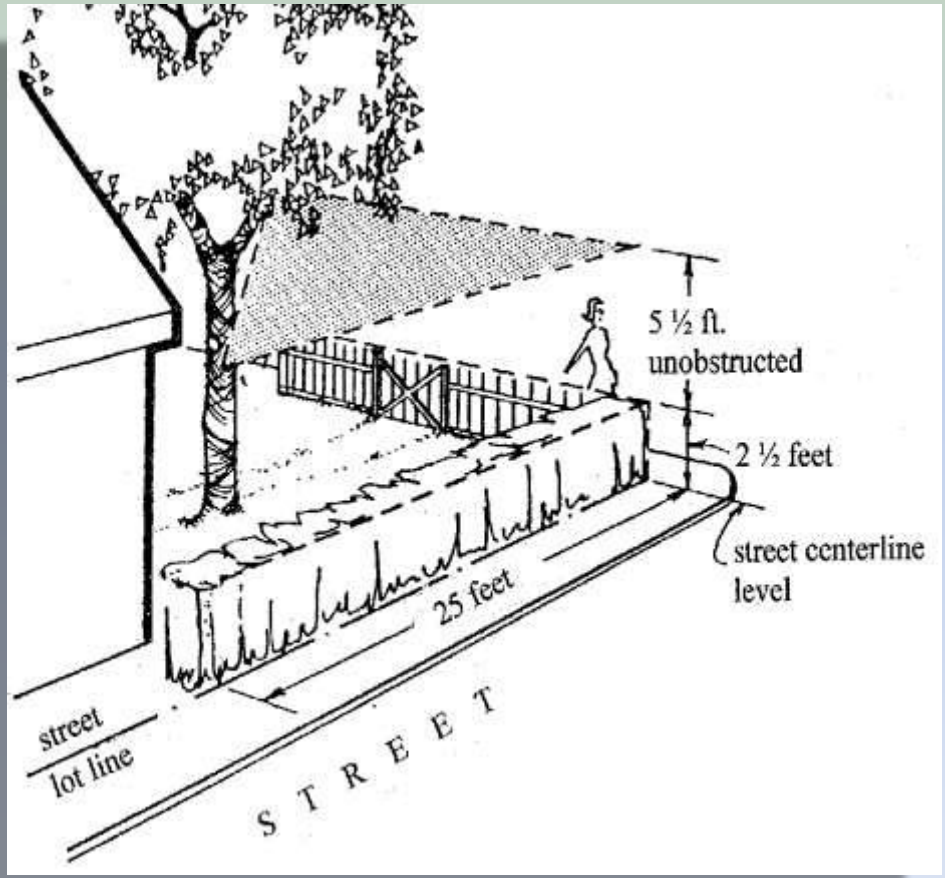
Importance Of Building By Bye -Laws

- **Building by laws are made to ensure orderly development of localities in urban areas so that every houses is assured of good ventilation and no building affects the lighting and ventilation of neighbors.**
- There has to be laws or regulations binding on the prospective builders, if not, the building constructed will be:
- 1.Un- Scientific
- 2. Un-healthy
- Inconvenient for the people to occupy.
- The buildings should not be constructed merely with profit motive without paying any attention to the health and comfort of the in - habitants.

Importance of Building By Bye -Laws

- The buildings must get **sufficient sunshine, air and ventilation.**
- **Open spaces** should be **well planned.**
- The **buildings** should **create better environment.**
- The **buildings should be located in healthy surroundings and should have an aesthetic appearance.**
- But to achieve all this, there has to be a suitable regulations or what are know as **model building bye laws, enforced strictly by the authorities, and followed by the builders honestly and truthfully.**

Importance of Building By Bye -Laws





Importance of Building By Bye -Laws

- These buildings Bye Laws Are Drawn Up By A Panel Experts In Various Branches Of The Building Industry Such As:
- **Town Planning**
- **Architecture**
- **Civil Engineering**
- **Electrical Engineering**
- **Air – Conditioning**
- **Fire- fighting and administration etc.**
- Theses building bye -
- laws when formulated are enforced on all buildings whether constructed by government, local bodies, private persons or agencies

Importance of Building By Bye -Laws





General Principles Of Building By B- Laws

- The building bye - laws are generally based on the following principles
- **The building bye - laws should be reasonably rigid and adequately flexible as they have to be sometimes revised according the improvements** affected in science and engineering and as per peculiar circumstances existing at the time.

General Principles Of Building By B- Laws

- These **laws should be advantageously** used for the common good of the people.
- **Minimum standards should be properly laid down and they should be strictly made** to be adhered to by all concerned.
- **Minimum floor space and cubic space** per member should be insisted upon.

General Principles Of Building By B- Laws

- The **size of any room should not be less than a specified minimum.**
- Taking into consideration the number of inmates in the **building the minimum window space and sanitary conveniences should be insisted upon.**
- At least **one window of the specified size/area should be provided in each room to open either on a street or open yard.**

General Principles Of Building By B- Laws

- The room should receive direct light and air from exterior open space on at least two sides to satisfy ideal conditions of air circulation.
- There should be some healthy relations between the cubic contents of the room building and open spaces around.
- The width and extent of the open space depends upon the height of the structure.

General Principles Of Building By B- Laws

- The height of the building is fixed as per the zone in which it is built.
- The width of the street should never be exceeded by the height of the building there.
- The openings admitting light and air should bear a prescribed ratio to the floor space.
- The set backs should be correctly followed

D.C.Rules

- The present development control regulations have come into force from 25 th march 1991. The regulations are framed to regulate the development /redevelopment in the Maharashtra region.

D.C.Rules





Applicability of Regulations

- These regulations shall apply to all development, redevelopment, erection and/or re-erection of a building, change of user etc. as well as to the design, construction or reconstruction of, and additions and alteration to a building.



Applicability of Regulations

- These regulations shall also apply to any revision of the development permissions/building permissions granted earlier under any Development Control Regulations .

Part Construction

- Where the whole or part of a building is demolished or altered or reconstructed, removed, except where otherwise specifically stipulated, these Regulations apply only to the extent of the work involved

Applicability of Regulations





Applicability of Regulations

Change of Occupancy / User:

- Where the occupancy or the user of a building is changed, except where otherwise specifically stipulated, these **Regulations shall apply to all parts of the building affected by the change.**

Applicability of Regulations

Reconstruction : -

- The **reconstruction in whole or part of a building which has ceased to exist due to an accidental fire, natural collapse or demolition,** having been declared unsafe, or which is likely to be demolished by or under an order of the **Municipal Council** or **Nagar Panchayat** and for which the necessary certificate has been given by the said **Municipal Council or Nagar Panchayat shall be allowed subject to the provisions in these Regulations**

Applicability Of Regulations





Applicability Of Regulations

Conflicts In Provisions -

- If **there is any overlapping of provisions or any conflicts between the existing provisions and the provisions** in the **Development Control & Promotion Regulations** for **Municipal Councils and Nagar-panchayats in Maharashtra**, then matter shall be referred to the Director of Town Planning, Maharashtra State, Pune whose decision shall be final.

Applicability Of Regulations

Applicability of CRZ Regulations -

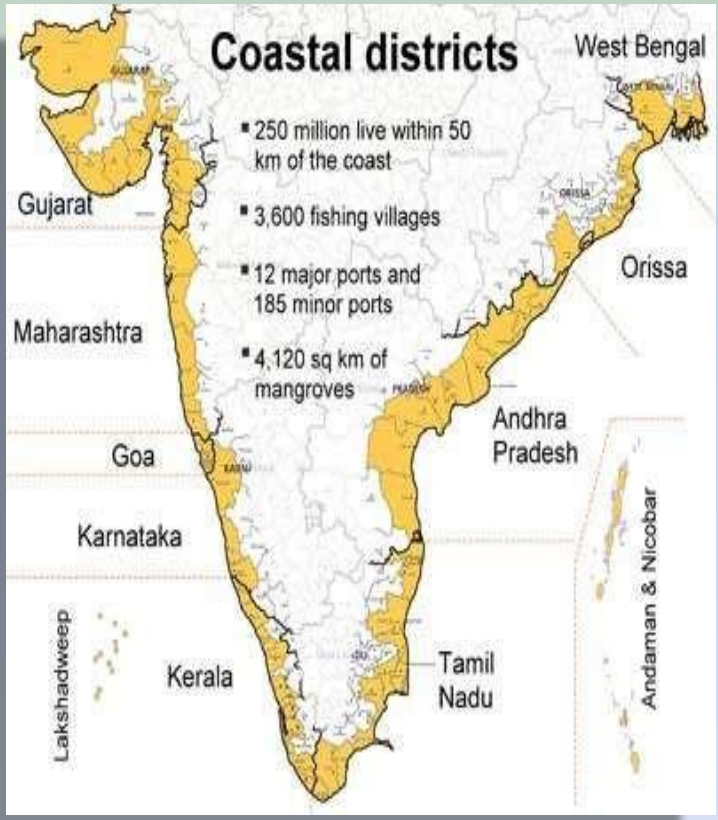
- Any development within CRZ areas shall be governed by the Coastal Regulation Zone Notification No.S.O.19(E), dated 6.1.2011 as amended from time to time, wherever applicable.

Applicability of Heritage Regulations -

- The heritage regulations, establishment of Heritage Conservation Committee and the list of Heritage Sites shall be applicable as previously sanctioned by the Government/concerned Competent Authority.

Applicability Of Regulations





Applicability Of Regulations





The Set Backs Should Be Correctly Followed

- The minimum distance between individual buildings should be rigidly controlled.
- Necessary water supply and sanitary connections should be made to every unit.
- There should be uniformity as regards drainage connections, water supply, gas and electricity.
- The buildings should have adequate fire-fighting arrangements if over 69 ft. in height

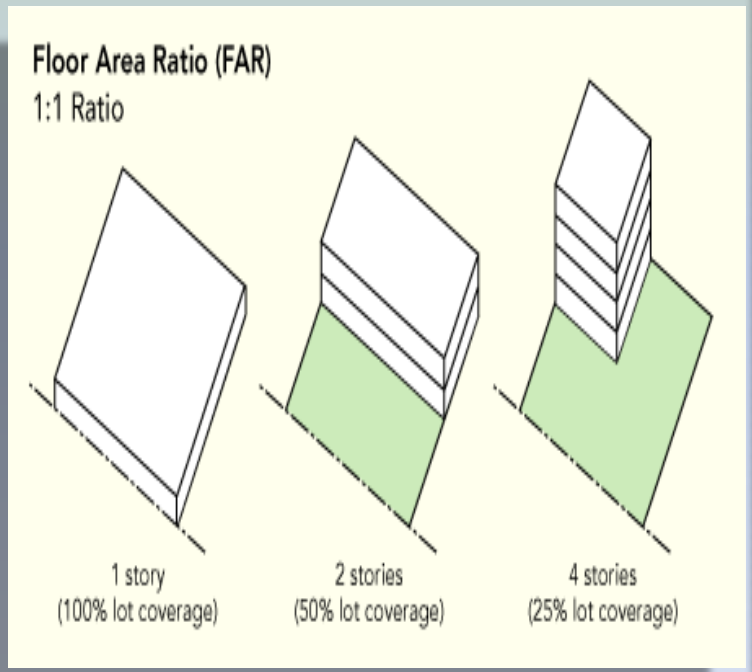
The Set Backs Should Be Correctly Followed

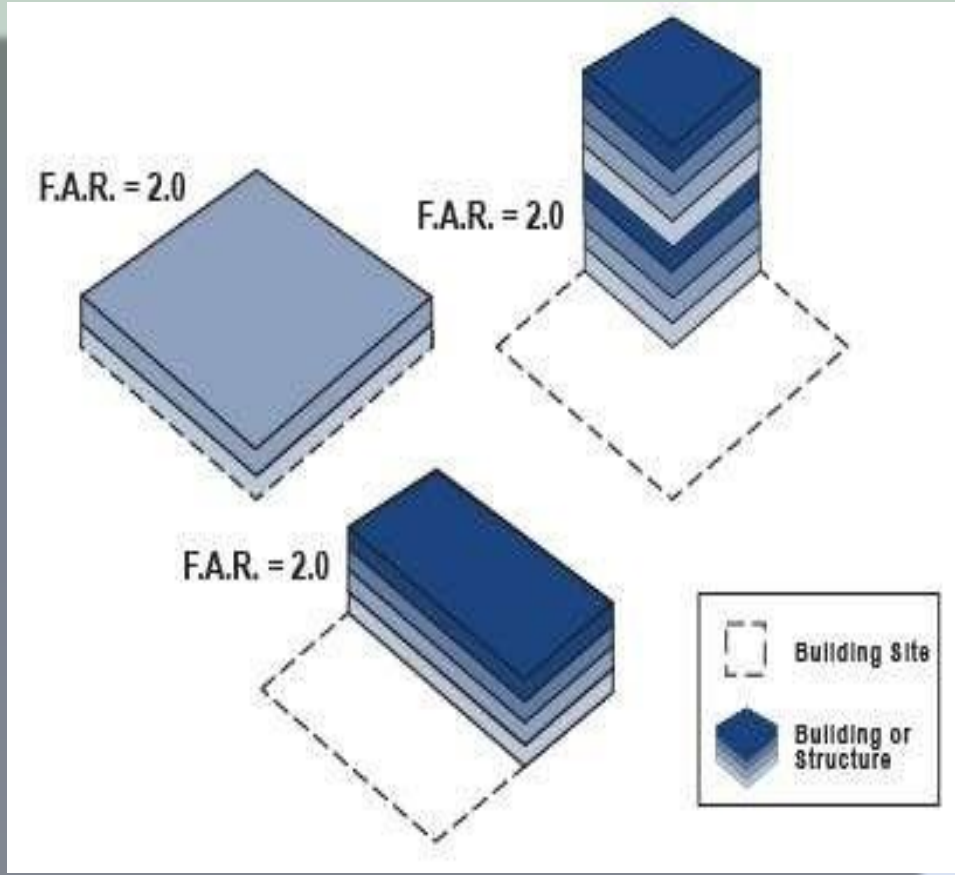
- In fact, it is essential that there should be control over the user of the buildings and control over the materials and construction of the building

The Set Backs Should Be Correctly Followed

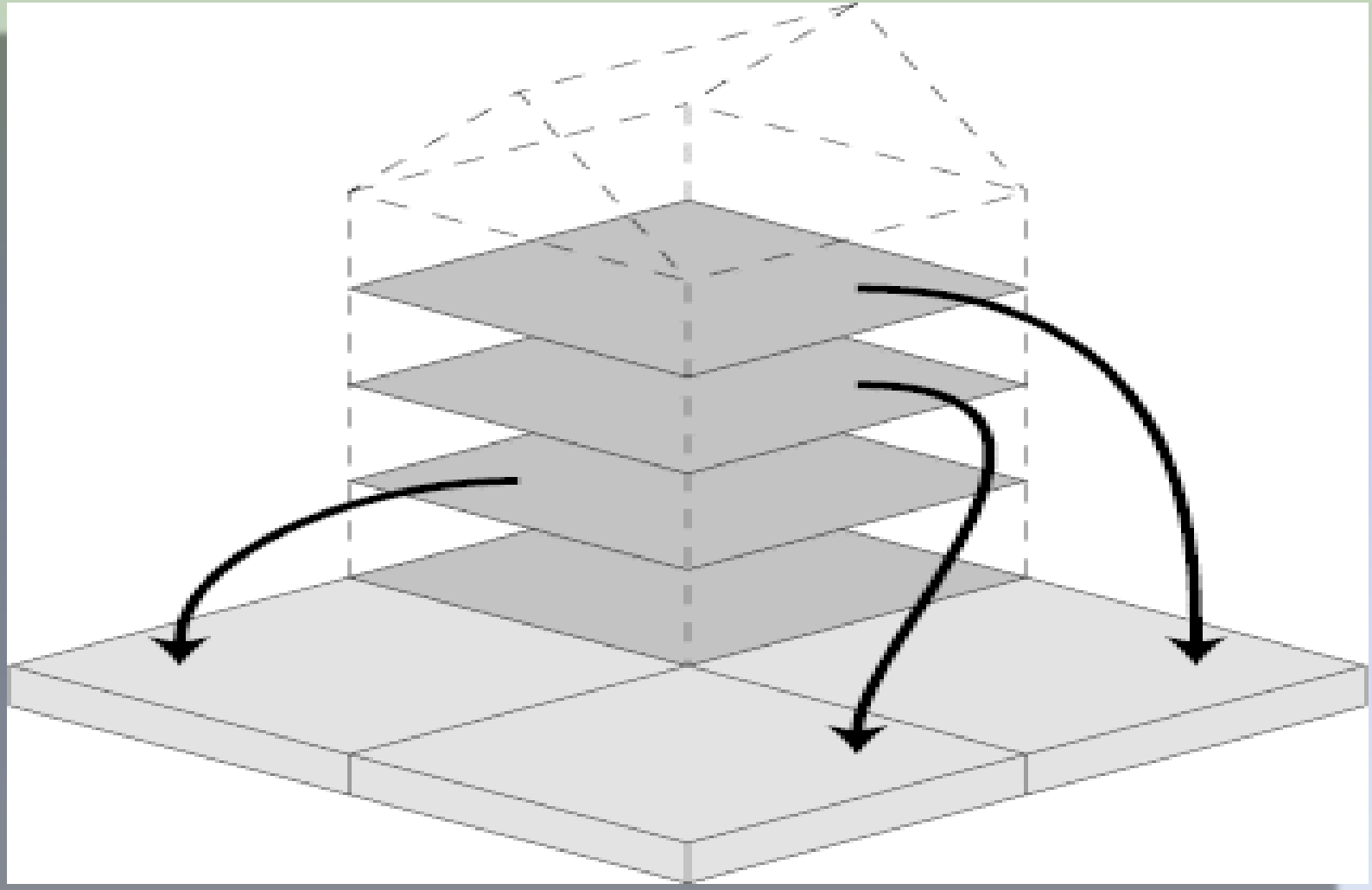
- **Floor Area Ratio** As a formula: **Floor Area Ratio** = (Total covered area on all floors of all buildings on a certain plot) / (Area of the plot) **The Floor Area Ratio (FAR) or Floor Space Index (FSI)** is the Ratio of the total floor area of buildings on a certain location to the size of the land of that location, or the limit imposed on such a ratio.
- **The Floor Area Ratio is the total building square footage (building area) divided by the site size square footage (site area).**
- Thus, an **FSI of 2.0 would indicate that the total floor area of a building is two times the gross area** of the plot on which it is constructed, as would be found in a multiple - story building

Floor Area Ratio





Floor Area Ratio

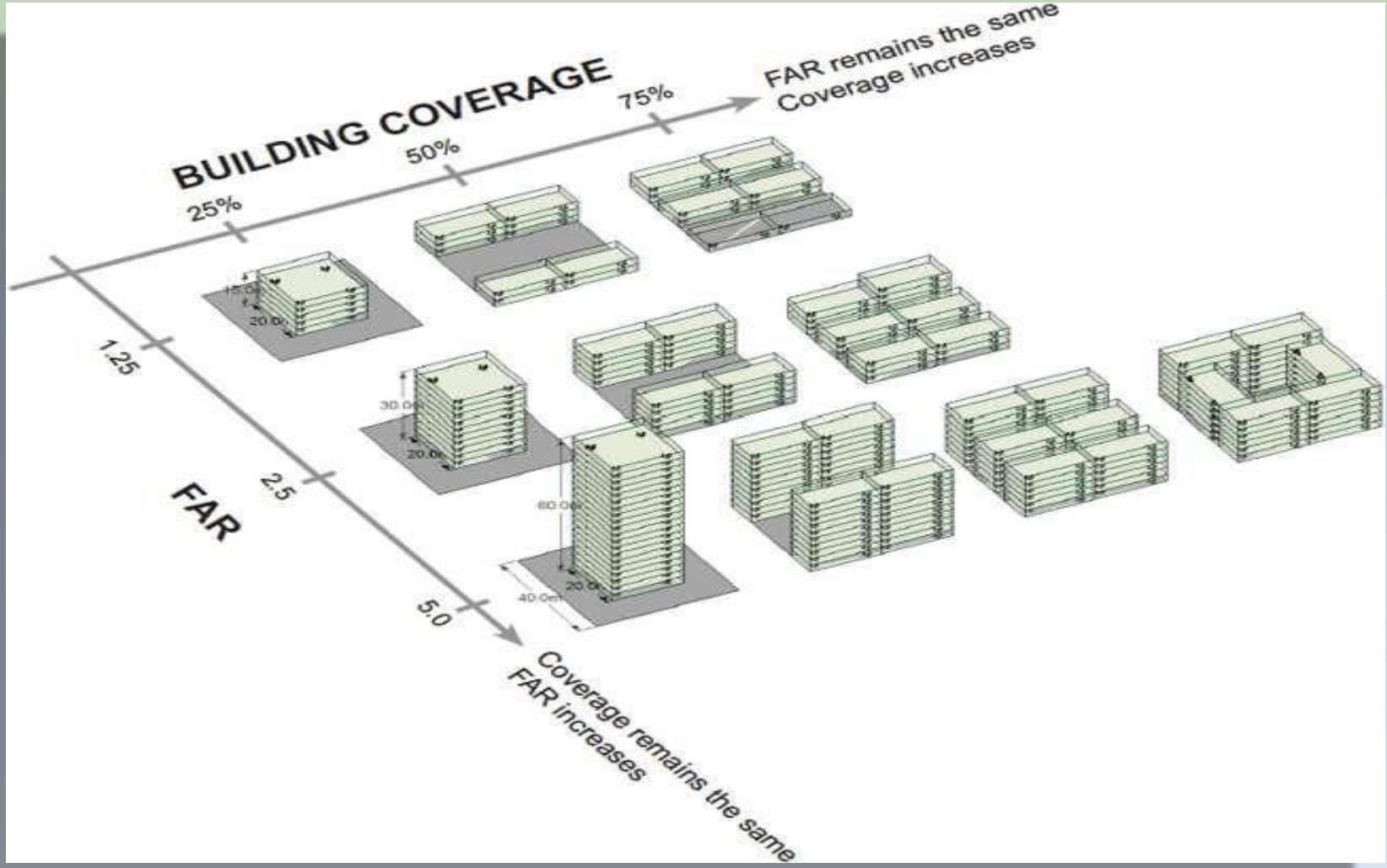


Floor Area Ratio

- **"Floor Area Ratio (FAR)"-** The quotient of the ratio of the combined covered area (plinth area) of all floors, excepting areas specifically exempted under these regulations, to the total area of plot, viz.: -

- $$\text{Floor Area Ratio (FAR)} = \frac{\text{Tota1 Covered Area on All Floors}}{\text{Plot Area}} \times 100$$

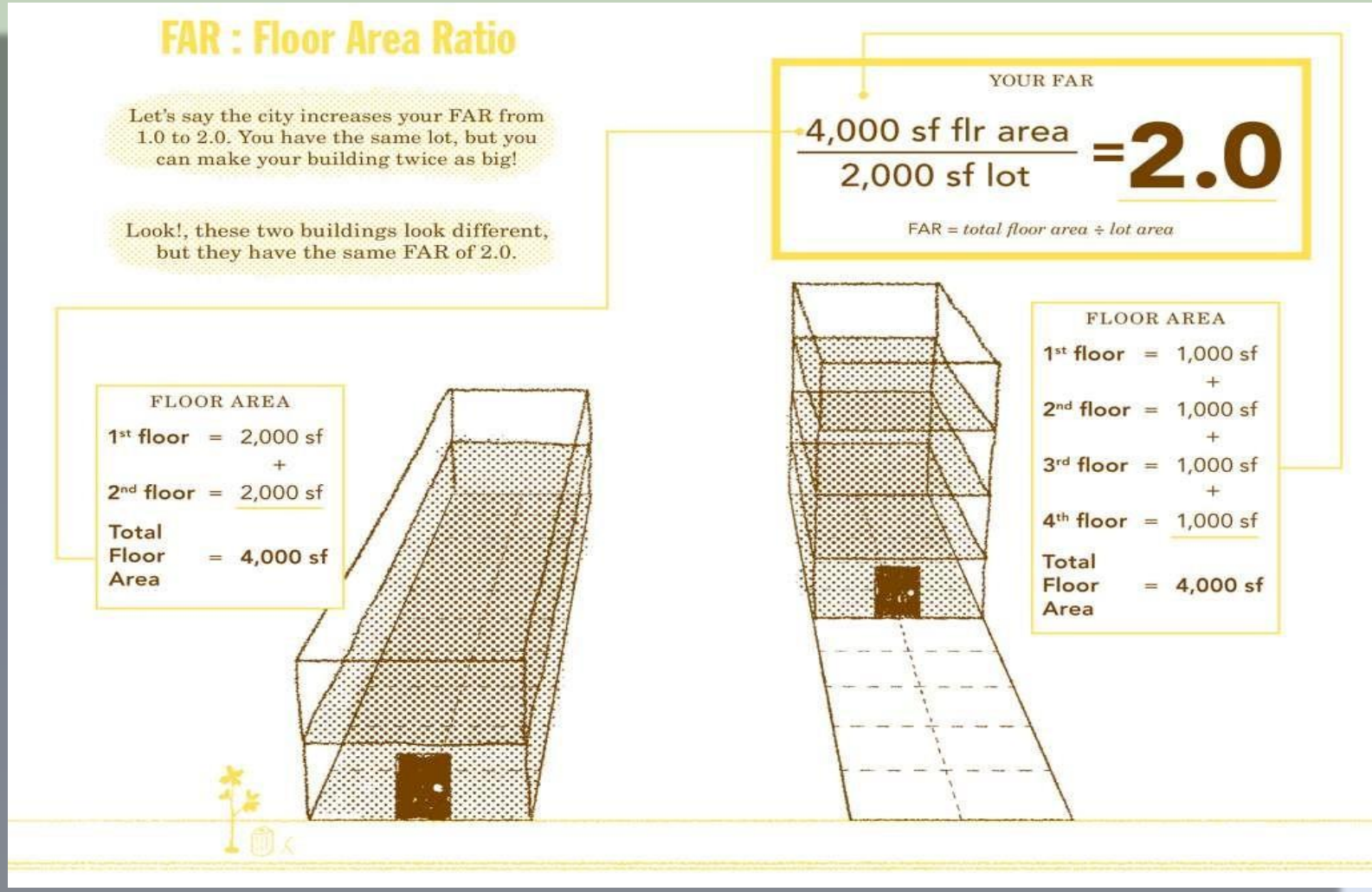
Floor Area Ratio



Floor Area Ratio

- Thus, FSI or is the upper limit to the built space you may construct on a given plot. **It is the ratio of allowed built-up area to the plot area.**
- For example, on a **10,000 ft²** plot which has an FSI of 2, you may construct a maximum of **20,000 ft²** of area, and no more.
- FSI can also be called FAR or .Certain types of spaces, such as basements, parking areas, and utility rooms, are, which means that they do not need to be counted in the FSI calculations.

Floor Area Ratio



Floor Area Ratio

- **The following are not counted in FSI calculations:**
  - Basements
  - Stilt Parking
  - Staircases
  - Lifts and lift lobby (lobby area to an extent equal to lift area, additional lobby areas are counted)

Floor Area Ratio

- Pump **rooms, utility areas, security cabins**
- **Shafts**
- **Society Office up to 12 m²** if there are less than 20 apartments, and 20 m² if more
- **Gymnasium up-to 2% of FSI area**
- One **Servants "toilet per floor up to 2.2 m²** with access from lift lobby Refuge Areas and terraces Normally, 15% of the plot must be reserved as a recreation area. If the plot area is greater than 2,500m², then this 15% is also subtracted from the total FSI of the plot.

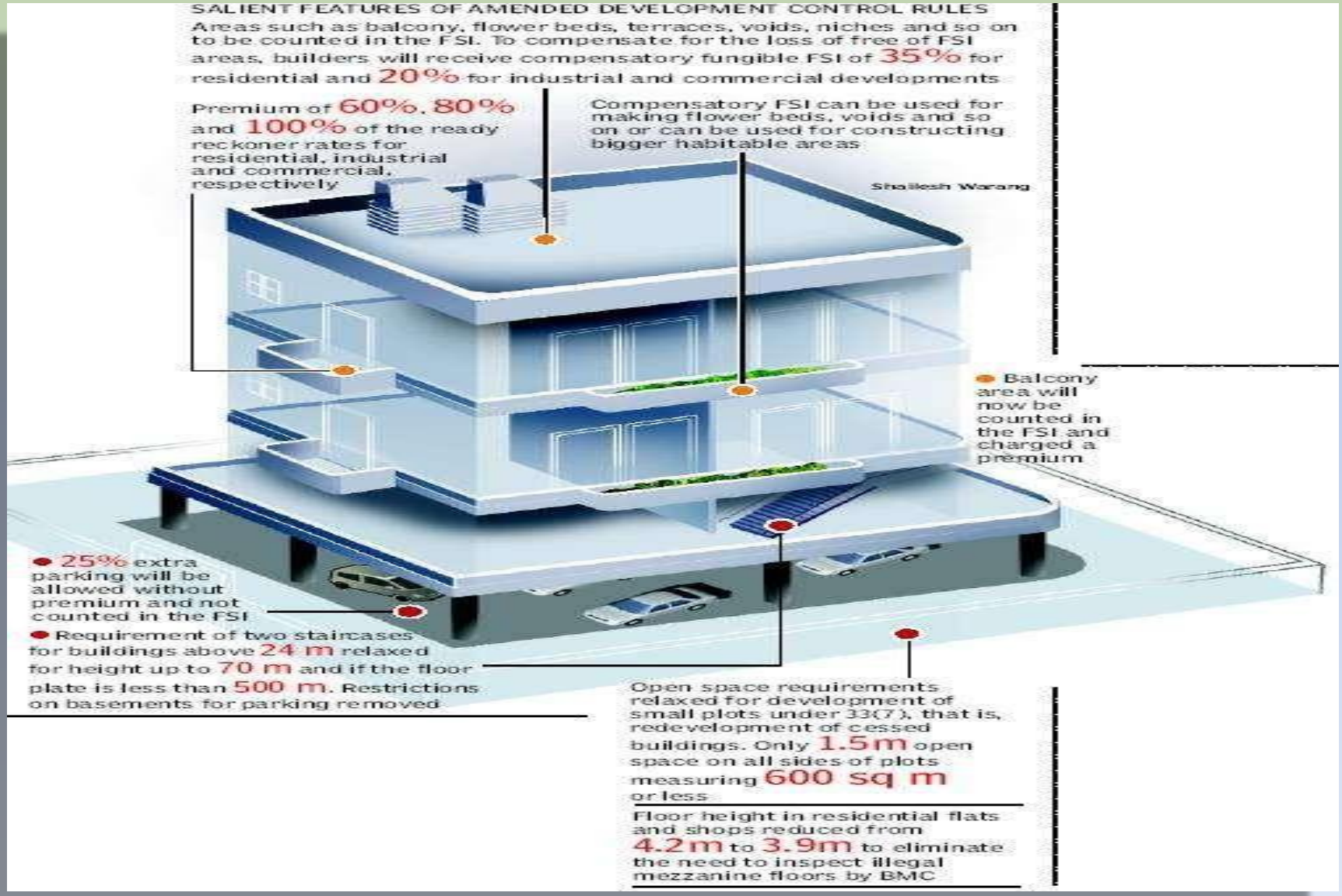
Fungible FSI

- The word **Fungible** from the Latin root describes something that acts as a replacement for something else. **The municipal corporation of Mumbai introduced this system to curb misuse of existing building regulations by developers.**

Fungible FSI

- Developers would build space over and above the **allowable FSI by the means of some grey areas in the building regulations.** These grey areas centered a round things that were free of FSI, or not counted in FSI calculations, such as flower beds, services shafts, and balconies. In the fungible FSI system, the allowable FSI on a plot is **increased by 35% ,** with a maximum cap on the total construction area, with no exemptions. **This serves to reduce 'overbuilding' on plots by developers.**

Fungible FSI



Fungible FSI

- **The fungible FSI seeks to act as a replacement for or a legalization of the misuse of regulations**, but with a clear mathematical limit that should not be exceeded. As of late 2013, fungible **FSI is applicable to all plots in Mumbai with the exception of those that fall under Coastal Regulation Zone (CRZ) limits.**

Fungible FSI

- The **Maharashtra Government**, in an attempt to provide balanced playing field to Developers and reduce arbitrary decision-making, has amended the **Development Control Regulations (DCRs) for the State Capital City, Mumbai.**
- According to the new **DCR Amendments, Balconies, Flower Beds, Terraces, Voids and Niches would now be counted in the Floor Space Index (FSI).** To compensate for the loss of free-of-FSI areas, Fungible FSI to the extent of 35 per cent for Residential Development and 20 per cent for Industrial and Commercial Developments has been allowed with premium.

Fungible FSI

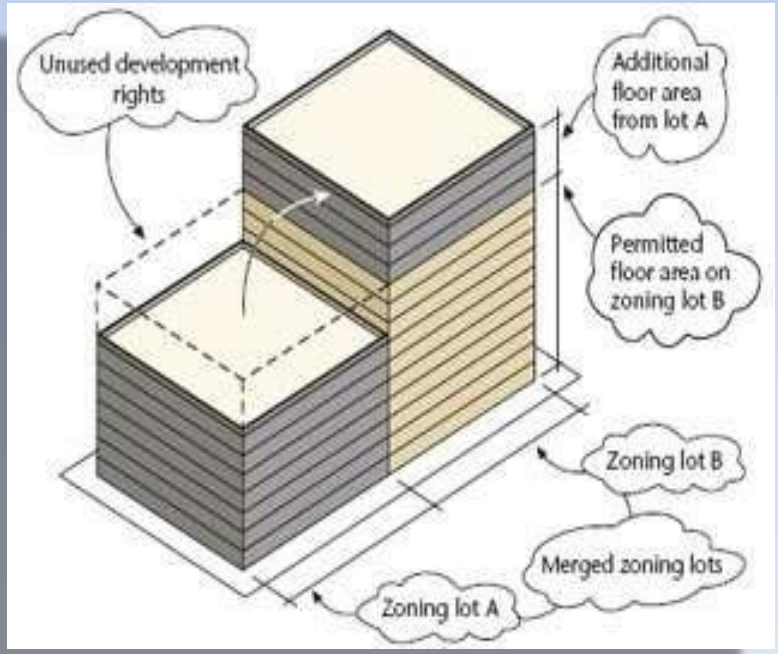
- **Fungible FSI would be available at 60 per cent premium for Residential, 80 per cent for Industrial and 100 per cent for Commercial at the Ready Recknor Rates (RR Rates)** which are revised from this January 1st, 2012 ranging between 5 per cent and 30 per cent in 716 zones of Mumbai. **Fungible FSI can be used for making Flower-Beds or Voids; else used for constructing bigger habitable areas.**

Fungible FSI

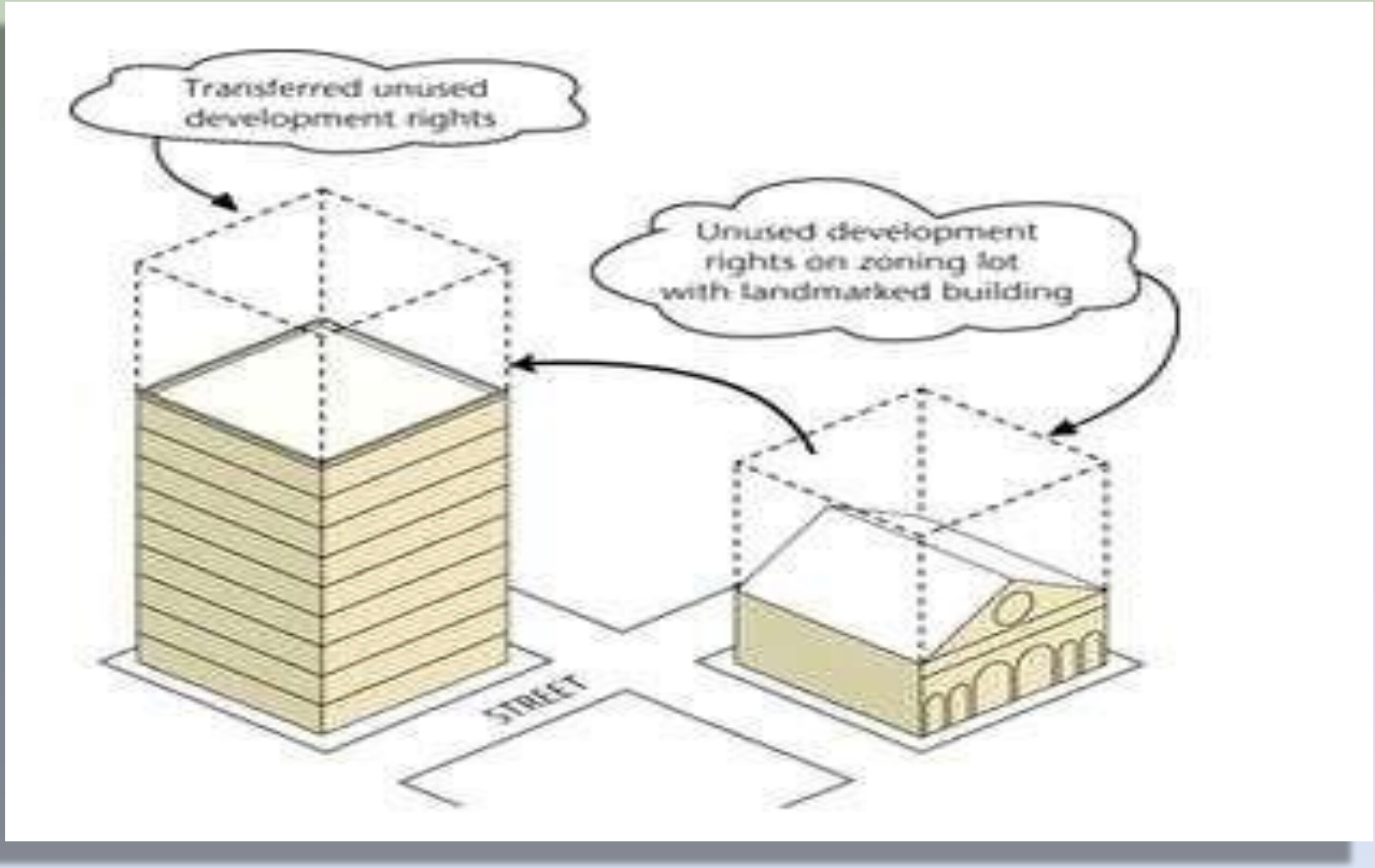
- However, to protect the interests of the existing owners and occupiers so as to avoid the transfer of **Fungible FSI** in respect of existing building to the free sale portion by the Developers, it has also been further clarified that the **Fungible FSI in respect of rehab portion would not be transferable to the free sale area of the Developer**. No premium for Fungible FSI would be charged for the members whose flats were being redeveloped though the space restrictions would be the same.
- The parking would be available as per the provisions of the DCR, but 25 per cent more at the option of the Developer. This would be without premium and without being counted in the FSI.

TDR

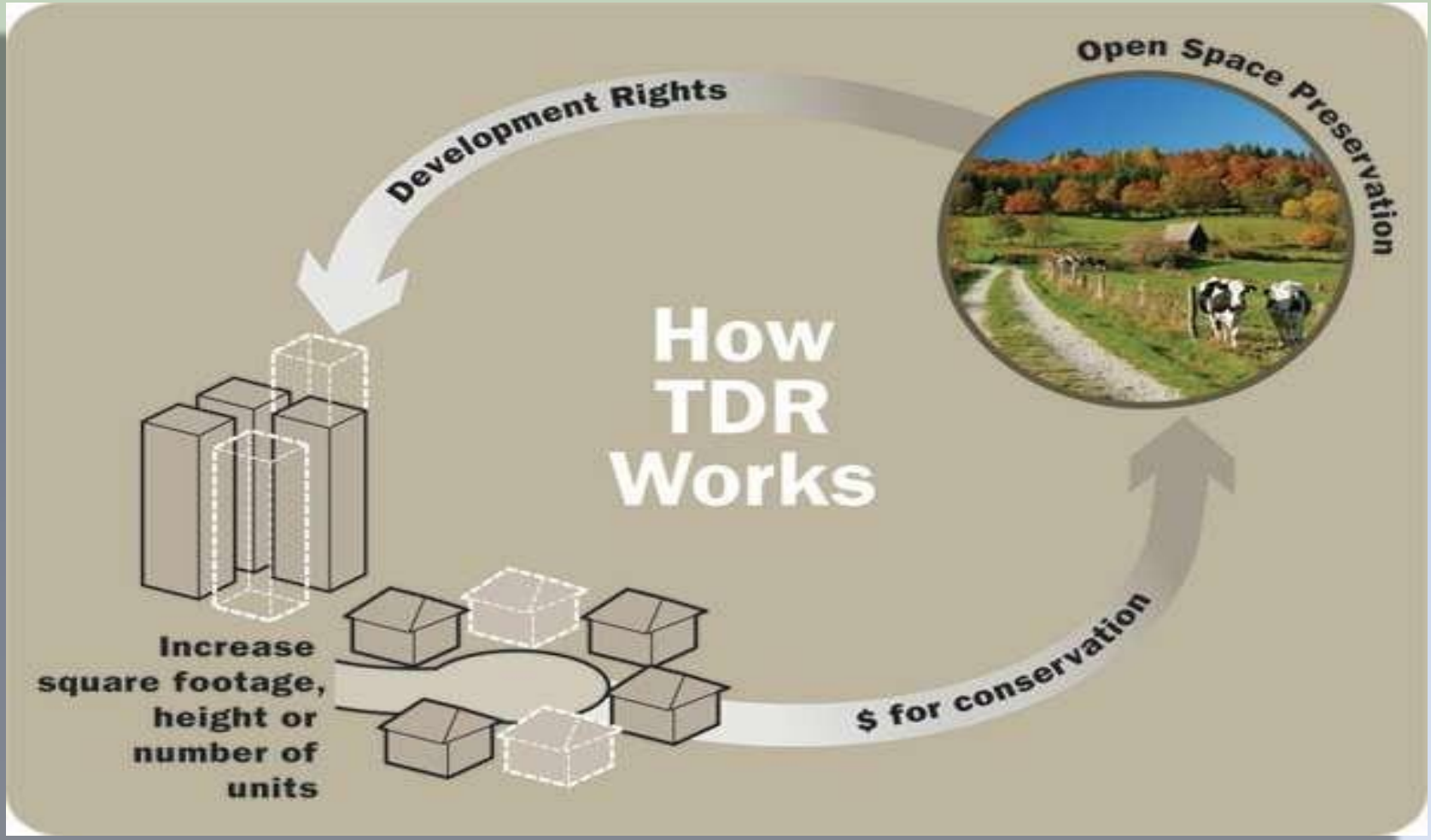
- **Transferable Development Rights** are a mechanism to reduce new construction **in crowded areas and shift it to less dense parts of the city.**



TDR


Transferred unused development rights
Unused development rights on zoning lot with landmarked building
STREET

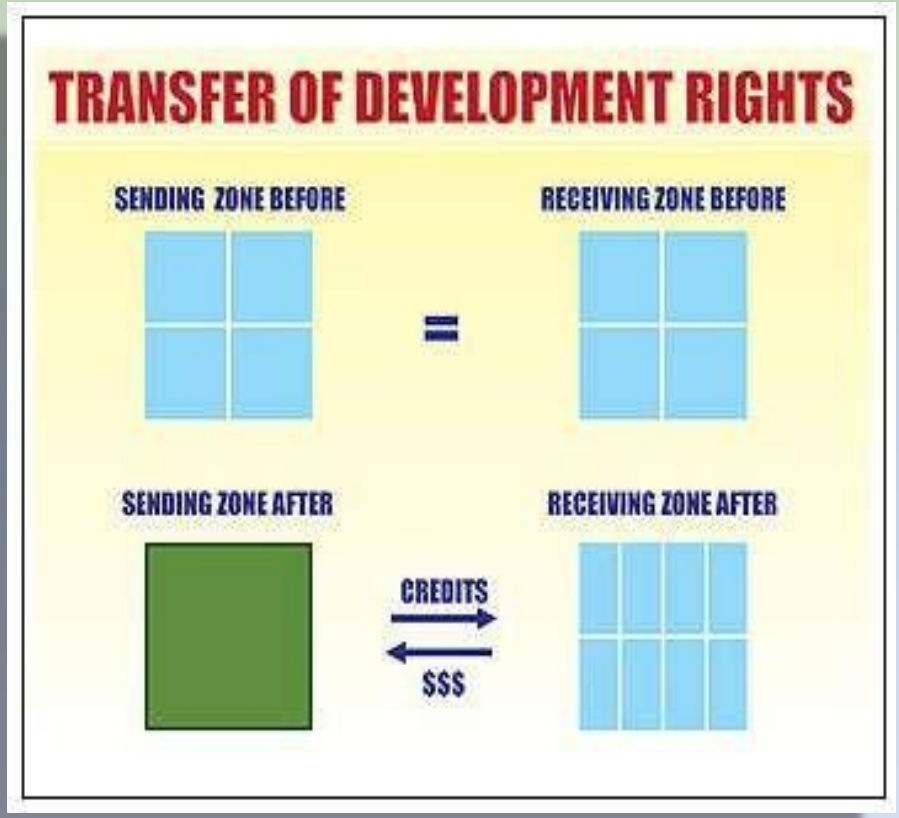
TDR


Development Rights
Open Space Preservation
How TDR Works
Increase square footage, height or number of units
$ for conservation

TDR

- In Mumbai, TDR was initiated to **prevent new construction in south Mumbai** and shift it northwards. It works as follows: say you own a 10,000 ft^2 plot with an **FSI** of 1.3 in South Mumbai, on which you have an existing building with 8,000 ft^2 of built-up area. To utilize your full **FSI** allowance, you would have to build 10,000 x 1.3 = 13,000 ft^2 of space. This means you have 5,000 ft^2 of extra capacity in your plot which you are not using. With TDR you can then **sell this right to build 5,000 ft^2 to someone north of you** . You then cannot build more than the existing 8,000 ft^2 on your plot.

TDR


TRANSFER OF DEVELOPMENT RIGHTS
SENDING ZONE BEFORE
RECEIVING ZONE BEFORE
=
SENDING ZONE AFTER
RECEIVING ZONE AFTER
CREDITS
$$$



TDR

- FSI in Mumbai city is 1.33 & FSI in the suburbs is 1.0.
- 2. in suburbs TDR potential is permissible with total cap of 2.0.

Norms For FSI Utilization For Different Kind Of Buildings

- Residential Use Zone The residential areas are developed either as

(a) Plotted development or

(b) Group housing/flatted development.

- **Density pattern** i.e. (high density, high medium density, low medium density or low density) are followed for working out the pattern of development with respect to the size of the plot, number of dwelling units on each plot, setbacks, FAR and the number of storeys/height of the building.

Norms For FSI Utilization For Different Kind Of Buildings





Norms For FSI Utilization For Different Kind Of Buildings

Buildings within the Residential Use Zone

- Buildings for various uses/activities within the residential use zone forming part of the residential layout plan are to be constructed with the norms of the coverage, FAR, height and others as applicable to that size of a residential plot.

Norms For FSI Utilization For Different Kind Of Buildings



Norms For FSI Utilization For Different Kind Of Buildings

- **Plotted Development** The layout plans for residential scheme are formulated keeping in view
- That there would be **sufficient light and air in the buildings when constructed**
- That there would **be protection against noise, dust and local hazards**
- That there would **be sufficient open space for various family needs**

Norms For FSI Utilization For Different Kind Of Buildings

- That the **circulation and access is easy and is safe from accident point of view**
- That, **as far as possible, the plots are of regular shape and size** and
- These **are logically arranged in a systematic manner so as to give a regular pattern of development in the form of row houses,** detached and semi-detached houses and if necessary the **regular bungalow type plots.**

Norms For FSI Utilization For Different Kind Of Buildings

Residential Premises –

- Following table is suggested for different size of the plots applicable, ground coverage, FAR, height and number of dwelling units for a residential area:

Sl.No.	Plot Area (sq.mt)	Maximum Ground Coverage %	FAR	No.of DUs.	Maximum Height (mt.)
Low-Income Group Housing (mainly for large cities/towns)					
1.	30	75	150	1	8
2.	Above 30 upto 50	75	150	2	8
Normal Housing (mainly for large, medium and small towns)					
3.	Above 50 upto 100	65	180	3	12
4.	Above 100 upto 250	65	180	3	12
5.	Above 250 upto 500	55	165	6	15
6.	Above 500 upto 1000	45	120	8	15
7.	Above 1000 upto 1500	40	100	8	15
8.	Above 1500 upto 2250	33 1/3	100	12	15

Group Housing

- The number of dwelling units are calculated on the basis of the density pattern given in the development plan, taking into consideration a population of 4.5 persons per dwelling unit.
- **Minimum size of the plot** 2250 sq m.
- **In hill towns** 5000 sq m.
- **Maximum ground coverage** 35%
- **Maximum FAR** 125 (higher FAR may be given depending on the pattern of development and should not exceed 150)
- **Maximum Height** 15 m. (for plot sixes up to 4000 sq m.) and 26 m. for plots above 4000 sq m.
- **In hill areas** 15 m. for all size of plots.
- **Number of dwelling units** To be calculated on the basis of the net plot area of a particular neighborhood. This may vary between 50 DUs. to 124 DUs. per ha.

Group Housing



Non- Residential Premises

Hostel

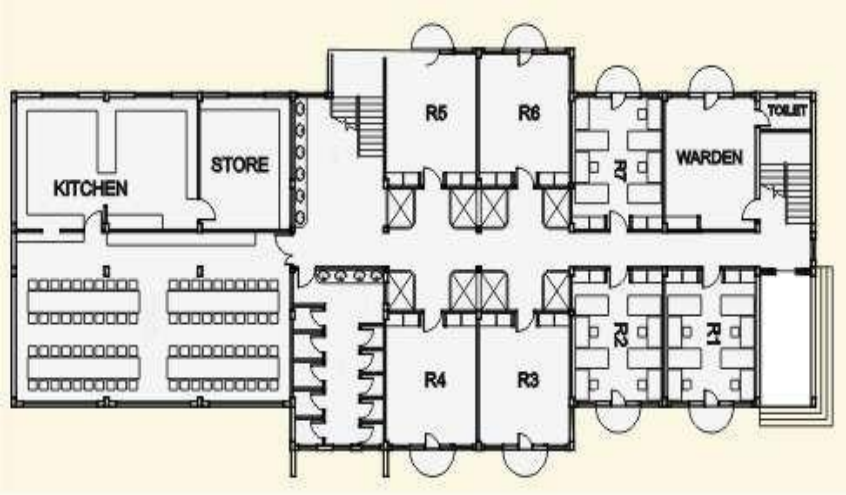
- **Maximum ground coverage** 33.33%
- **Maximum floor area ratio** 100
- **Maximum height** 26mt.

Other Control

i) Minimum R/W in front 12 m.

ii) Basement up-to the building envelope to the maximum extent of 50% plot area shall be allowed and if used for parking and services should not be counted in FAR.

Non- Residential Premises





Non- Residential Premises

Guest House, Boarding House and Lodging House

- **Minimum plot size** 500 sq m.
- **Maximum ground coverage** 33.33%
- **Maximum floor area ratio** 100
- **Maximum height** 26 m.
- **Other Controls:**
- i) Minimum R/W in front 20 m.
- ii) Basement up-to the building envelope to the maximum extent of 50% of plot area shall be allowed
- and if used for parking and services should not be counted in FAR.

Non- Residential Premises

Convenience Shopping

- Maximum ground coverage 40%
- Maximum floor area ratio 60
- Maximum Height 15 m.
- In hills 6 m.

Non- Residential Premises





Non- Residential Premises

Local Shopping

- **Maximum ground coverage** 30%
- **In hills** 35%
- **Maximum FAR** 100
- **Maximum Height** 15 m.
- **In hills** 9 m.

Community Centre

- Maximum ground coverage 25%
- In hills 30%
- Maximum FAR 100
- Maximum Height 26 m.
- In Hills 15 m.

Non- Residential Premises

Hospitals

- Minimum plot size 6000 sq m.
- Maximum ground coverage 25%
- Maximum floor area ratio 100
- Maximum Height 26 m

Other Controls

- Area to be used for housing of essential staff is indicated in the norms for health facilities. In such an area the regulations of group housing shall apply.
- Basements below the ground floor and to the extent of ground coverage shall be allowed and if used for parking and services should not be counted in FAR.

Non- Residential Premises



Non- Residential Premises

Health Centre/Nursing Home

- Maximum ground coverage 33.33%
- Maximum floor area ratio 100
- Maximum height 15 m.
- Basement shall be as in case of Hospital

Non- Residential Premises

a) **Filling-cum-service station size** 36 m. x 30 m. and 45 m. x 33 m.)

- i) **Ground Coverage** 20%
- ii) **FAR** 20
- iii) **Max. Height** 06 m.
- iv) Canopy Equivalent to permissible ground coverage within setback line.
- v) **Front Setback Min.** 06 m.

Non- Residential Premises



Non- Residential Premises

b) Filling Station Size

(30 mt. x 17 mt. and 18 mt. x 15 mt.)

- i) **Ground Coverage** 10%
- ii) **FAR** 10
- iii) **Max. Height** 06 m.
- iv) Canopy Equivalent to permissible ground coverage within setback line.
- v) Front Setback Min. 03 m

Non- Residential Premises

Other Regulations

- i) Shall be approved by Explosives/Fire Dept.
- ii) Ground coverage will exclude canopy area.
- iii) Mezzanine if provided will be counted in FAR
- iv) Wherever the plot is more than **33 m. x 45 m.** development norms shall be restricted to as applicable to the size i.e. 33 m. x 45 m. both in urban and rural areas.

Non- Residential Premises

Compressed Natural Gas (CNG) Mother- Station

- i) **Plot Size (Max.)** 36 m. x 30 m.
- ii) **Maximum Ground Coverage** 20%
- iii) **Maximum Height** 4.5 m. (Single Storey)
- iv)Building Component Control room/office/Dispensing room, store, pantry and W.C.

Compressed Natural Gas (CNG) Mother-Station



Educational Facilities

Nursery School

- Maximum ground coverage 33.33%
- Maximum floor area ratio 66.66
- Maximum height 8 m
- In hills 6 m

Educational Facilities





Educational Facilities

Primary School

- Maximum ground coverage 33%
- Maximum floor area ratio 120
- Maximum Height 15 m

Educational Facilities

Higher Secondary School

- Maximum ground coverage 30%
- Maximum floor area ratio 120
- In hills 100
- Maximum Height 15 m

Higher Secondary School





Educational Facilities

College

- Maximum ground coverage 25%
- Maximum floor area ratio 100
- In hills 75
- Maximum height 15 m.

Note:

- In case of the above premises the total area of the plot shall be divided in
- **i) School/college building area**
- **ii) Play field area**
- **iii) Parking area**
- **iv) Residential and hostel area**
- The maximum ground coverage and FAR shall be calculated only on the areas meant for building

Educational Facilities





FSI Norms For Redevelopment Across Mumbai

- In Mumbai, the permitted **FSI varies in various locations depending upon the land and type of existing residence whether a Slum, Tenanted, MHADA,** Cluster or a Co-operative Society and also the projects to be redeveloped.



FSI Norms For Redevelopment Across Mumbai

- As per **Development Control Rules 33(7) of Mumbai**, in the case of **Cess Building**, every tenant who is having less than 300 sq ft carpet area shall be given minimum of 300 sq ft ownership flat and the tenants having **300 to 753 sq. ft area, the existing area.** Any area above 753 sq. feet will not be granted to the tenants.
- Of course, the **maximum ceiling was 2.5 Floor Space Index (FSI) under this category of buildings**. Many of the old buildings in areas were not becoming viable under the above rules and therefore, the government took a decision to increase the same to **3 FSI.**

FSI Norms For Redevelopment Across Mumbai



FSI Norms For Redevelopment Across Mumbai

- In case of redevelopment of plots under the above category, the builder gets incentive FSI of 50% or 60% of FSI over and above the FSI consumed to re-house the existing tenants.
- In such cases FSI goes beyond 6 or 7. There is no limit. For MHADA layout, the FSI is 2.5.
- In case of projects approved by MMRDA in Mumbai Metropolitan Region other than Mumbai under Rental Housing Scheme, the FSI allowed is 4

FSI Norms For Redevelopment Across Mumbai





Reforms in D.C rules by U.D department of state government and municipal authorities.

- In India, the **combined effect of multiple layers of poorly conceived central, state and municipal regulations contribute to an artificial urban land shortage.** As a result **urban land prices are abnormally high in relation to India's household income,** and households consume less floor space than they could afford if the regulatory environment were reformed.

Reforms in D.C rules by U.D department of state government and municipal authorities.

- In addition, some regulations have a negative impact on the spatial structure of cities. **By unreasonably reducing the amount of floor space that can be built in centrally located areas**, and by making land recycling difficult, some regulations tend to "push" urban development toward the periphery.
- By making **urban development financially unfeasible in areas where there is high demand for commercial** or residential space, **some regulations encourage corruption.**

Regulations should be regularly audited and submitted to a cost benefits analysis

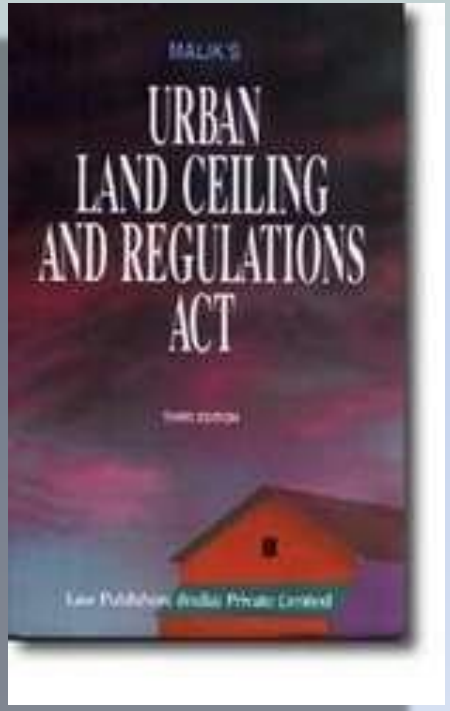
- In reviewing or auditing land regulations it is convenient to divide them according to their impact on markets.
- Some regulations contribute to a decrease in land supply, other artificially increase **land consumption and therefore demand for land.**
- The **double effect of restricting supply** and mandating high land consumption has an evident impact on price.

Reforms in D.C rules by U.D department of state government and municipal authorities.

- Government regulations or practices contributing to a decrease in land supply:
- *Urban Land ceiling act*
- The major effect of the urban land ceiling act has been to freeze large areas of land in legal disputes. These areas are not available for development or redevelopment. An additional negative impact of the act was to prevent private developers to assemble land for subsequent development. The act gave a de facto monopoly on land development to government developers.

Reforms in D.C rules by U.D department of state government and municipal authorities





Reforms in D.C rules by U.D department of state government and municipal authorities.

Rent control

- The effect of rent control on the supply of new rental stock is obvious and well documented. However rent control laws have also an effect on land supply and city shape. Rent control contributes to a decrease in land supply because buildings which are under rent control cannot be redeveloped or even renovated.

Reforms in D.C rules by U.D department of state Government and Municipal authorities.





Reforms in D.C rules by U.D department of state government and municipal authorities

Regulations preventing or slowing down the conversion of land from one use to another.

- Any change of use, even when approved by Master plans, requires lengthy approval to become effective. This is particularly serious at the periphery of cities where land has to be converted from agricultural to urban use.

Reforms in D.C rules by U.D department of state Government and Municipal Authorities







Reforms in D.C rules by U.D department of state government and municipal authorities

Master Plans Ignoring real estate demand

- **Master plans allocate land between various uses and limit the amount of floor space which can be built on specific parcels**, either directly through maximum FSI or indirectly through set backs, plot coverage ratio, and maximum number of floors. **While these types of control are not objectionable per se, the parameter used are often arbitrary and have been set without taking into account the efficiency of city structure or the affordability of different social groups.**

Reforms in D.C rules by U.D department of state government and municipal authorities





Reforms in D.C rules by U.D department of state government and municipal authorities

High Stamp Duty

- High stamp duties discourage land transactions, and as a consequence reduce the supply of land on the market. High stamp duty incites to grossly under-declare the real value of land. This in turn adversely affects the possibility of using land as collateral for construction financing.



Reforms in D.C rules by U.D department of state government and municipal authorities

Large Institutional land holdings

- **Government entities or parastatals such as Railways often own large tracts of land in cities.** Because this land cannot be sold on the market to the benefit of the owning institution, **it is often underused, or used in a way incompatible with its real market value.**

Reforms in D.C rules by U.D department of state government and municipal authorities





Reforms in D.C rules by U.D department of state government and municipal authorities

Very low Property Taxes

- **Very low property taxes and property taxes based on actual rents rather than on land values create an incentive to hold vacant** or underused land, thus decreasing the amount of land on the market.



Reforms in D.C rules by U.D department of state government and municipal authorities

Very Low Property Taxes

- **The failure to provide primary infrastructure with a capacity consistent with demand is often cited as a justification for constraining development intensity, in particular low FSI.** It is important to realize that an adjustment of land use regulation to actual market demand will also require the provision of primary infrastructure of sufficient capacity.

Expected Impact Of Land Regulatory Reforms

- **The careful review and reforms of the regulations mentioned above would result in a lower cost for urban development** and for housing. An additional benefit will be a more efficient spatial organization for cities.

Expected Impact Of Land Regulatory Reforms



Expected Impact Of Land Regulatory Reforms

- More compact cities, more efficient land use. **No enclaves of under use or unused land; more efficient use of existing primary infrastructure.**
- **Increase share of the housing stock supplied by unsubsidized formal private sector developers,** decrease in illegal subdivisions and slum areas.
- Generally **lower land prices but higher prices in some prime commercial and business areas.**

Expected Impact Of Land Regulatory Reforms

- Decrease in trip length due to more compact cites and because of more intense use of land in the CBD (**Central business district)** (less dispersion of employment). **A more intensely used CBD allows a better efficiency of transit and therefore should increase urban air quality in the long run.**
- **Increase in the consumption of floor space per person for both residential and business use.** This should result in an increase of welfare for households and an increase in productivity for firms.

Expected Impact Of Land Regulatory Reforms





Expected Impact Of Land Regulatory Reforms

- Average urban population densities are likely to stay constant as more efficient land use and higher FSI are likely to be balanced by a higher floor consumption.



Expected Impact Of Land Regulatory Reforms

- Finally, by reducing the difference between what is allowed and what is financially feasible, **land use reform should reduce significantly the opportunity for corruption**





References

- **Urbanization Urban Development & Metropolitan Cities in India**
  - **Dr V. Nath** Concept Publications
- *Standardized Development Control And Promotion Regulations For Municipal Councils And Nagar Panchayats In Maharashtra.*
- *Building Bye-laws Municipal Building Bye - Laws And Restriction In Designing Of Buildings*
- Internet Websites

GHT

CC BY